

Module 41 - Array Higher Order Methods (forEach vs map)

1. What are Array Higher-Order Methods?

Array Higher-Order Methods are built-in JavaScript methods that take a callback function as an argument to perform operations on array elements.

2. What is forEach()?

The forEach() method is used to iterate over each element of an array and perform an operation. It does not return a new array.

3. What is map()?

The map() method is used to iterate over each element of an array, perform an operation, and return a new array with the modified values.

Example Code:

I. forEach() Method

```
let fruits = ["Apple", "WaterMelon", "MuskMelon", "Banana"];
// fruits.forEach(printFruit);
// function printFruit(currentElement, index, totalArray){
//   console.log(currentElement, index, totalArray);
// }
// Executes the callback for each array element.
// forEach() always returns undefined.
let newArr = fruits.forEach((cElement)=>{
  console.log(cElement.toUpperCase());
  return cElement;
});
console.log(newArr);
```

II. map() Method

```
// fruits.map(function(currentEle, index, totalArr){  
//   console.log(currentEle, index, totalArr);  
// })  
  
// Creates and returns a new array.  
  
let newArr1 = fruits.map((cElement, index)=>{  
    return {id : index + 1, fruit : cElement};  
});  
  
console.log(newArr1);
```

III. Chaining Method

```
// let newArr3 = fruits.forEach(cEle => cEle.toUpperCase()).sort().fill("123");  
  
// console.log(newArr3); // TypeError: Cannot read properties of undefined (reading 'sort')  
  
// map() returns a new array, so methods can be chained.  
  
let newArr2 = fruits.map(cEle => cEle.toUpperCase()).sort().fill("123");  
  
console.log(newArr2);
```

IV. Conditional Based Statement

```
// Returns a new array of boolean values.  
  
let newArr4 = fruits.map((cEle) => {return cEle == "Apple"});  
  
console.log(newArr4);  
  
// Executes the callback for each element but does not return a new array.  
  
let newArr5 = fruits.forEach(val => console.log(val == "Apple"));
```